

Night Sky Explorer VR

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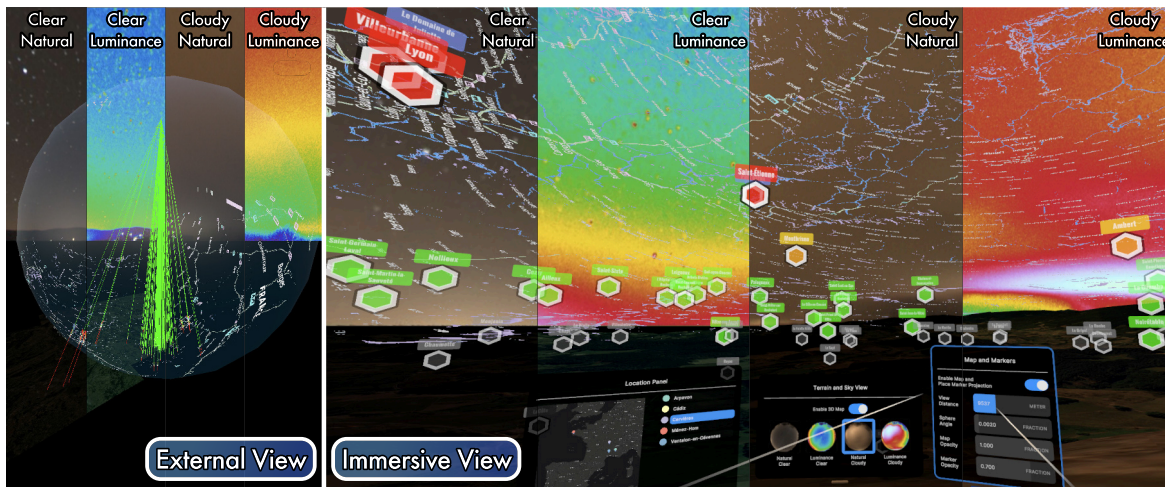


Figure 1: Example views of a location being shown with 3D geometry, and stereographically projected road networks and color-coded place names, divided into an external view (left) and an immersive view with the user interface panels (right). Each view is interleaved with the four available all-sky images for this location: natural and luminance view of a clear or a cloudy sky.

ABSTRACT

We present a Virtual Reality application that immersively visualizes nighttime skyglow and its sources of light pollution. Utilizing Unity with Cesium and OpenStreetMap geospatial data, the system integrates calibrated all-sky images processed with the Sky Quality Camera software to provide natural and luminance views of the night sky. By projecting road maps and location markers onto a surrounding sphere using inverse stereographic projection, users can intuitively explore the correlation between observed skyglow and its sources. This tool offers educational and exploratory potential, highlighting the impact of artificial light pollution on the night sky in an immersive and realistic context.

Index Terms: Light pollution, Skyglow, Immersive visualization, Geovisualization, Sky Quality Camera, Virtual Reality.

1 INTRODUCTION

Artificial light at night (ALAN) adversely affects astronomical observations, ecosystems, and human health [9]. Measuring and monitoring light pollution, visualizing its impact on the night sky, and identifying pollution sources are essential for education and mitigation [1, 11, 3]. Traditional methods rely on static images and separate maps [6], which lack immersion and intuitive understanding. We propose a VR application that combines calibrated all-sky images with immersive 3D geospatial data, allowing users to

explore and comprehend light pollution sources in a realistic and augmented context. The application aims to be useful in raising awareness surrounding the issue and consequences of ALAN in a more immersive and impactful way, and make it easier for people to grasp the issue's scale.

2 SYSTEM OVERVIEW

The application is developed in the Unity engine¹ using OpenXR and aimed at high-resolution VR HMDs such as the Meta Quest 3 and Apple Vision Pro. It integrates Cesium² for photorealistic 3D data tiles from Google Earth [7] and road maps, and OpenStreetMap's Overpass API³ for location markers, with a custom immersive visualization interface.

Users can select from a set of available locations captured and provided by DarkSkyLab [2] shown on a map with all-sky nighttime images produced by Sky Quality Camera (SQC, Euromix Ltd, Ljubljana, Slovenia), a specialized software that processes images obtained with a calibrated digital camera [6], providing both natural and luminance views (see Fig. 1) in a complete 360-degree dome panorama. The luminance view from SQC displays sky brightness levels using a specific and intuitive color scale [8]. SQC also offers the option to isolate artificial skyglow by subtracting celestial bodies like the Milky Way or bright stars [5]. A custom shader renders the all-sky images as the skybox.

Since the images are dome-shaped and not full spherical panoramas, we incorporate photorealistic 3D via the Cesium plugin. The terrain is illuminated in real time by the chosen skybox, enhancing realism and providing contextual grounding in the VR environment.

Places such as cities, towns, and villages that likely contribute to skyglow are retrieved via Overpass API "node around" queries

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¹Unity Technologies: Unity Engine

²Cesium GS, Inc.: Real-World 3D Geospatial Capability for Unity

³Overpass API — OpenStreetMap Wiki

and are rendered as billboarded marker objects that always face the viewer with the same apparent size. Their scale, color, and distance visibility are fine-tuned and controlled based on their “place” key in the OpenStreetMap data.

3 INVERSE STEREOGRAPHIC PROJECTION

To visualize the potential sources of light pollution corresponding to observed skyglow, we project the road maps and location markers (e.g., villages, cities) onto a surrounding sphere using the inverse of the stereographic projection. This method is based on multi-perspective geovisualization that aims to create a focus+context view through gentle geometric deformation [10].

Fig. 2 illustrates the principle: a sphere is created around the viewer, with its vertical axis passing through the viewer. From the sphere’s zenith, rays are cast to the ground and the markers on it. The markers are projected where these rays intersect with the sphere’s surface. The same principle applies to maps made of vectors and bitmap textures.

The projection causes the map and markers to curve upwards from the landscape, surrounding the user. Markers that would be below the horizon or behind hills become visible, and their vertical position indicates their distance from the viewer — objects projected higher up are further away. Projection parameters are adjustable in real-time, allowing users to account for different geographies and desired effects, such as stronger compression of distances or better separation of locations (Fig. 1 and Fig. 2).

The inverse stereographic projection of the road map tiles is implemented using a Unity shader graph, enabling real-time adjustments of the sphere’s radius and vertical position. Since these parameters are not intuitive at first in their effect on the projection, an abstraction is provided to the user in the form of two more directly relevant controls. Their effects are illustrated in Fig. 2:

- View Distance: Sets the distance at which the sphere intersects with the ground plane. The angle of the intersection stays constant.
- Sphere Angle: Adjusts the angle at which the sphere intersects the ground plane, measured in the *intersection ratio* from tangential contact (0, the sphere barely touches the plane) to a dome overhead (0.5, the sphere is bisected). The view distance stays constant.

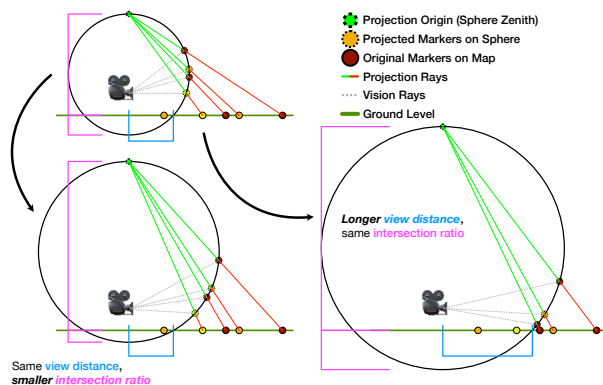


Figure 2: External view of the stereographic projection with different parameters for view distance and intersection ratio.

4 RESULTS AND DISCUSSION

Prototype versions have been demonstrated to experts in urban light pollution, including to members of a scientific consortium dedicated in part to studying artificial light at night, and to employees

of a specialized company devoted to the scientific study of light pollution and its impacts on ecosystems. Early feedback indicates that the tool could be very helpful for educational and exploratory purposes. Users can visually correlate bright areas in the sky with specific sources of light pollution at various distances — a common task when analyzing the results of measurement sessions.

By immersively visualizing both luminance and natural color photographs, the application highlights the impact of artificial light pollution and aims to facilitate a deeper understanding of its sources. The intuitive controls and immersive environment might enhance user engagement compared to traditional methods.

5 FUTURE WORK

A planned user study will evaluate the application’s effectiveness in educational and analytical settings compared to traditional visualization methods, providing quantitative data on its potential impact and guiding further improvements to the user interface.

Future enhancements in functionality include integrating radiance maps from VIIRS-DNB satellite data in addition to road maps and markers to show the radiance of areas as seen from space. Another planned feature is to generate new all-sky images directly from VIIRS-DNB data, allowing exploration of regions without photographic data and validating simulation results against existing all-sky photographs [4]. Building on this, simulating VIIRS-DNB data itself would also enable urban planners and stakeholders to explore different planning scenarios with their direct impact on the night sky, such as shielded lighting or reduced brightness, to identify optimal strategies for reducing light pollution.

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